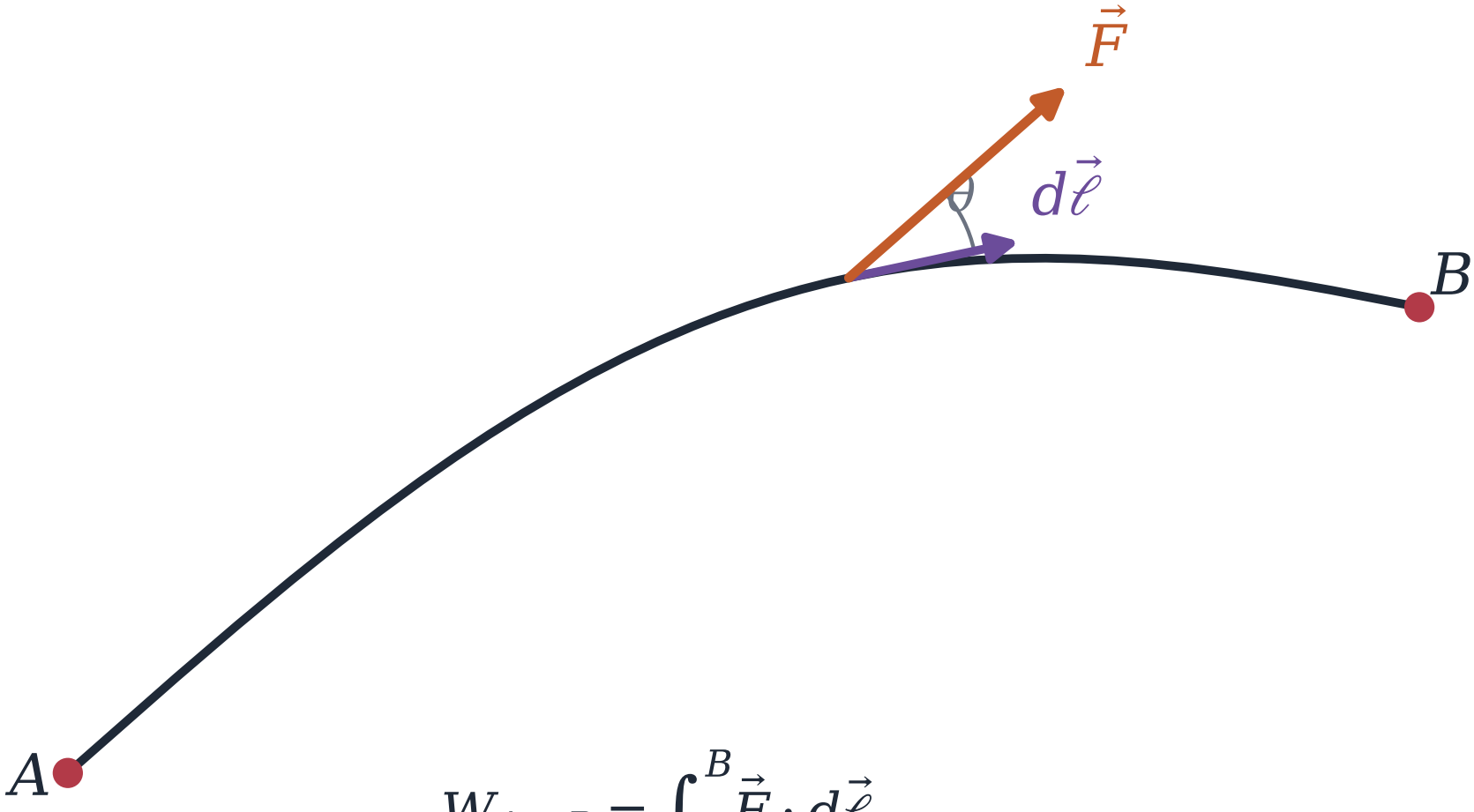

$$W_{A\to B} = \int_A^B \vec{F} \cdot d\vec{\ell}$$